

My Research Journey

If you've made it this far, thank you for taking the time to explore my website and learn a little more about my work. Here, I outline my research journey: from my first experience conducting research as an undergraduate student, to the work I am doing today, and where I hope to take my research in the future.

Biological Anthropology and a Holistic Approach to Human Health

Before you begin, I would like to address a question that I have been asked countless times throughout my PhD:

“Why Anthropology?”

Because my research relies heavily on statistical genetics and computational methods, many people often assume it belongs solely within genetics or epidemiology. Even I have occasionally simplified my research by leaving out the word "Anthropology" altogether. But the more I reflect on this question, the more I realize that the confusion is not about my research, but rather a common misconception of what anthropology is and the questions it seeks to answer.

This preface, therefore, aims not to justify why my work belongs in anthropology, but to explain why anthropology provides the academic home for the questions that motivate this dissertation. Although my methods are rooted in statistical genetics, the questions I ask are anthropological.

Anthropology is the study of humanity, seeking to understand what makes us human by examining our biology, culture, evolution, and society across time and space. Anthropology includes many subfields, including biological anthropology. Biological anthropologists have long sought to understand *why humans vary*. Historically, this question was explored through skeletal morphology, comparative anatomy, physiology, and population studies. Today, advances in genomics, computational biology, and statistical genetics provide unprecedented opportunities to investigate these same questions at the molecular level. With these advancements, we can ask questions central to biological anthropology, such as: How do humans vary? What biological and environmental processes generate that variation? And how does that variation influence health?

Large-scale genomic studies sampling human populations worldwide have enabled us to understand variation in disease resistance or susceptibility between and within populations, as well as to identify genetic risk factors for certain common diseases and disorders. Understanding population-specific biology is needed to develop therapies, medicines, vaccines, and treatments. Genetics in anthropology is ultimately intertwined with evolutionary biology, population genetics, and biological anthropology, allowing us to better understand human evolution, variation, and adaptability. O'Rourke (2003) noted that the modern approach to anthropology, intersected with molecular biology and bioinformatics, has gained the analytical and laboratory tools to dissect the molecular and genetic bases of human variation, a traditional focus within biological anthropology. The addition of genome scanning and linkage analyses has contributed to the growth of genetic epidemiology and the mapping of genes involved in complex phenotypes, helping us better understand humans.

However, genetics is not the only key factor, as anthropologist Paul Baker (1997) notes; any phenotype is influenced by a mixture of genes, innate behaviours, contributions from the physical environment (e.g., natural resources, climate, etc.), and the sociocultural environment (e.g., norms, values, learned behaviours). Baker deems this the “biocultural approach” to human phenotypes. Additionally, holism focuses on the whole person, where various aspects of biological and cultural life are of interest. Therefore, we cannot fully understand what it means to be human by studying a single aspect of our body or society. Using a holistic approach, anthropologists ask how different aspects interact and influence one another (Namie 2023).

Together, these concepts show that humans are not the product of either nature or nurture but a complex, nuanced mix of both. Our underlying biology and the environmental factors we create shape who we become. This key concept recognizes that we cannot understand humans by examining a single facet of biology alone. Human health is not determined solely by our genes or solely by our environment. Instead, it emerges from interactions among biological, behavioural, social, environmental, and evolutionary processes. This framework examines biological data (i.e., genetics) alongside psychosocial, lifestyle, and environmental factors to help us understand disease variability, etiology, progression, and outcome. Therefore, by using large-scale, real-world datasets and statistical genetics, we can study these factors as they relate to and interact with one another, rather than examining biology in isolation.

This perspective is central to my approach to research. Human health is not determined solely by our genes or solely by our environment. Rather, it emerges from interactions among biological, behavioural, metabolic, lifestyle, social, environmental, and evolutionary processes. My research therefore examines genetic variation alongside modifiable traits and behaviours using large population-based datasets and statistical genetic approaches. Rather than reducing disease to an individual variant or risk factor, I want to understand relationships among systems and the pathways through which they may influence human health.

Where It Started: My Undergraduate Research

My research journey began with an interest in forensic science. As a student fascinated by forensic biology and genetics, I pursued a BSc with a combined specialization in Biology and Forensic Science at Laurentian University. My undergraduate coursework introduced me to molecular biology and genetics. Still, research allowed me to move beyond classroom concepts and apply them to a real biological question.

In Dr. Nkongolo's laboratory, I conducted an undergraduate research project using simple sequence repeats, or short tandem repeats (STRs), to identify species-specific markers for red maple, silver maple, and their hybrid, Freeman's maple. I used wet-laboratory techniques including PCR, DNA quantification and purification, gel electrophoresis, and Sanger sequencing. The project aimed to develop genetic markers to distinguish species and support applications in reforestation and conservation. This experience introduced me to developing a research question, generating and interpreting genetic data, and communicating scientific findings.

Although I enjoyed the laboratory component of molecular genetics, the project also made me increasingly interested in human genetics and in the computational approaches required to interpret increasingly complex molecular datasets. This realization became an important turning point in my academic trajectory and ultimately led me toward biomedical informatics.

From Wet-Lab Biology to Computational Biology: My Master's Research

I pursued an MSc in Biomedical Informatics at Queen's University to develop the computational skills needed to work with large-scale biological data. In Dr. Amber Simpson's Medical Computing Lab, I completed a thesis focused on quantitative imaging phenotypes derived from CT images and tissue microarrays from cancer patients. My research investigated whether imaging and tissue-based features could identify tumours characterized by adaptive immune responses.

This work represented a major shift in my research approach. Rather than relying primarily on laboratory techniques, I worked with complex quantitative datasets. I applied supervised and unsupervised machine-learning and statistical methods, including principal component analysis, k-means clustering, random forests, support vector machines, linear regression, and Cox regression. The project taught me how computational methods can transform high-dimensional biological information into interpretable phenotypes and clinically relevant questions. It also strengthened my interest in integrating different types of biological data rather than studying each source independently.

My undergraduate and master's experiences established two foundations that continue to shape my research: an interest in genetic and biological variation, and an appreciation for computational approaches that allow us to investigate that variation at scale.

Expanding the Question: My PhD Research

These interests brought me to the University of Toronto, where I am pursuing a PhD in Evolutionary Anthropology with a Collaborative Specialization in Global Health, with research based at the CAMH Krembil Centre for Neuroinformatics. My doctoral research has expanded from studying individual forms of genetic variation toward understanding how genetic variation contributes to complex human traits, behaviour, cardiometabolic health, and brain health.

A central theme of my dissertation is that many genetic and non-genetic factors shape complex human phenotypes. I therefore use complementary approaches to move from identifying genetic associations toward understanding shared genetic architecture and potential causal relationships.

Current Research: From Genetic Variation to Brain Health

Project 1 — Tandem repeats and subcortical brain volume. My first doctoral project examines exome-wide associations between tandem repeats and subcortical brain volume. Tandem repeats are an underrepresented form of genetic variation compared with the single-nucleotide variants that dominate conventional GWAS. This project asks whether variation in tandem repeats contributes to differences in brain structure, extending my longstanding interest in repeat variation from my undergraduate work to human brain phenotypes.

Project 2 — Tandem repeats, alcohol use, and related behaviours. My second project investigates exome-wide tandem repeat associations with alcohol use and related behaviours. This extends the genetic questions from brain structure to behavioural phenotypes and reflects my broader interest in the relationship among genetic variation, behaviour, and health. Together, these projects allow me to investigate whether understudied forms of genetic variation contribute to complex traits relevant to both human behaviour and brain health.

Project 3 — From association to causal inference. My third project uses Mendelian Randomization to investigate potential causal relationships between tandem repeats, alcohol-related behaviours, and brain volume. While association studies can identify relationships between genetic variants and phenotypes, MR provides a framework for asking whether genetically proxied exposures may causally affect outcomes. This represents an important progression in my research, moving from describing genetic associations to evaluating possible mechanisms and causal relationships.

Project 4 — Shared genetic architecture of cardiometabolic health and Alzheimer's disease. My fourth project uses Genomic SEM to examine the shared genetic architecture of cardiometabolic traits, lifestyle-related factors, and Alzheimer's disease. Rather than examining traits individually, I am interested in whether groups of related phenotypes reflect shared latent genetic factors and how these factors relate to Alzheimer's disease. This work brings together my interests in complex traits, metabolic health, brain health, and multivariate statistical genetics, and represents the broader systems-level perspective that increasingly characterizes my research.

What Connects My Research?

Although these projects span different phenotypes and methods, they share a common question: how does biological variation contribute to differences in human health and behaviour, and how can we understand those relationships within the broader context of human lives? My research has progressed from identifying genetic variation to studying its association with complex phenotypes, to examining shared genetic architecture and causal relationships. At the same time, my research questions have become increasingly interdisciplinary, bringing together genetics, epidemiology, neuroscience, anthropology, and public health.

My research journey also shows an unexpected continuity. I began my undergraduate research studying short tandem repeats as molecular markers in maple species. During my PhD, I returned to tandem repeats, this time asking whether they contribute to variation in human brain structure and behaviour. The biological system and questions have changed dramatically, but the underlying curiosity about genetic variation has remained. This continuity reminds me that research trajectories do not always develop as a straight line; previous questions and skills can reappear in new contexts and become the foundation for discoveries.

Where I See My Research Going

Beyond my PhD, I hope to develop a research program at the intersection of genetics, brain health, aging, behaviour, and cardiometabolic health. One direction I am particularly interested in pursuing is the role of modifiable lifestyle and metabolic factors in brain aging and neurodegenerative disease. Alcohol use, obesity, blood pressure, lipid metabolism, glucose regulation, and related traits do not operate independently, and understanding their shared genetic and biological architecture may provide a more complete picture of pathways leading to brain health or disease.

I also hope to continue investigating forms of genetic variation that are often overlooked in conventional GWAS. Tandem repeats have been a particularly useful entry point into this area, but advances in sequencing and computational methods are creating opportunities to study a broader range of genetic variation. Understanding how these variants influence complex phenotypes could help address gaps left by SNP-focused approaches to human genetics.

Another future direction is to integrate genetic, molecular, behavioural, and environmental information more fully. I am interested in moving beyond questions of whether a variant is associated with a phenotype toward questions of why that association exists, how multiple traits influence one another, and under what environmental or behavioural contexts genetic effects may differ. This includes continued development of causal inference, multivariate genomic methods, and integrative analyses of genetic, transcriptomic, metabolomic, and phenotypic data.

Finally, I hope my future work will retain the anthropological perspective that brought these questions together. Human health is inherently contextual. Genetic susceptibility exists within environments, behaviours, cultures, and social systems, and these contexts can shape whether biological risk is expressed. My long-term goal is therefore not simply to identify genetic predictors of disease, but to contribute to a more comprehensive understanding of human variation and health by examining how biology interacts with the worlds in which people live.

Selected Conceptual References

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